

ESG Integration In Project Management

Project Principal And Practice Management Course MS(PM)

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Background

- In today's global environment, organizations are increasingly expected to operate responsibly, ethically, and sustainably.
- Climate change, resource depletion, social inequality, and governance failures have heightened the need for project-based industries to reassess how projects are planned, executed, and evaluated.
- Traditional project management historically emphasized the "iron triangle" of cost, time, and scope. However, modern project environments demand broader accountability that includes environmental stewardship, social responsibility, and strong governance structures.



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ESG Framework Integration



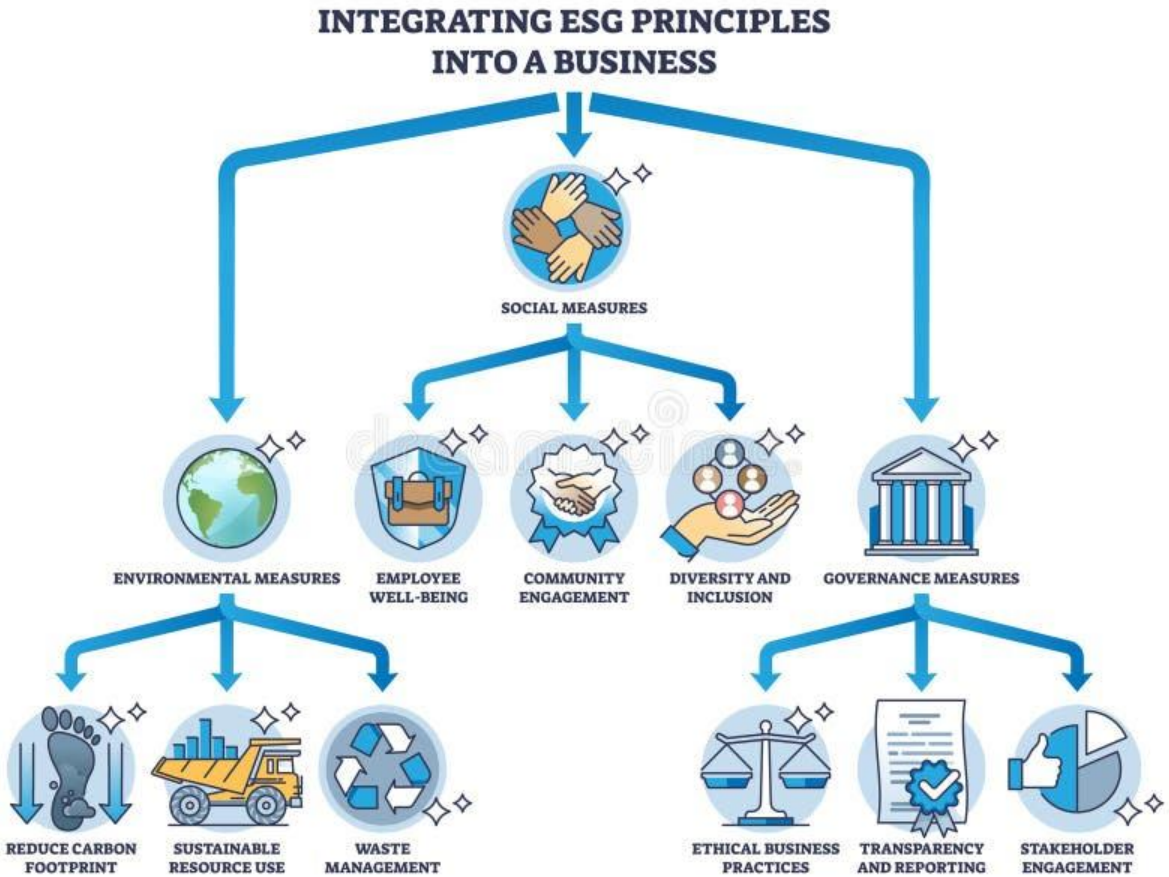
- This shift has led to the integration of Sustainability and **ESG (Environmental, Social, Governance)** frameworks within project management practices.

- International standards such as the UN Sustainable Development Goals (SDGs), Paris Climate Agreement, and corporate ESG reporting requirements have pushed organizations to adopt sustainability indicators across all project phases.

- As a result, project managers now must incorporate resource efficiency, carbon reduction strategies, ethical labor practices, stakeholder inclusion, and transparent governance mechanisms.

- Sustainability and ESG integration not only supports long-term ecological and social well-being but also enhances project resilience, risk mitigation, and organizational competitiveness.

- Therefore, understanding how ESG principles influence project decision-making, stakeholder management, and project outcomes has become a critical area of study in advanced project management programs.



Significance

Addresses Global Sustainability Challenges

This study highlights how project management can directly contribute to solving global issues such as climate change, environmental degradation, and social inequity through ESG-driven practices.

Enhances Project Performance and Long-Term Value

Integrating sustainability strategies can lead to improved lifecycle performance, cost savings, risk reduction, and greater long-term project value.



Supports Regulatory and Compliance Requirements

Governments, investors, and international bodies now require ESG reporting. This study helps organizations align with global standards, certifications, and compliance frameworks.

Significance

Strengthens Organizational Reputation and Stakeholder Trust

Projects that demonstrate responsible environmental, social, and governance practices gain stronger stakeholder support, public trust, and investor confidence.

Provides a Framework for Future Project Managers

As sustainability becomes a core expectation, this study equips future project managers with the knowledge and tools needed to lead sustainable, ethical, and resilient projects.

Promotes Innovation and Sustainable Design Thinking

Exploring ESG integration encourages creative solutions, green technologies, and innovative methodologies that transform traditional project delivery.

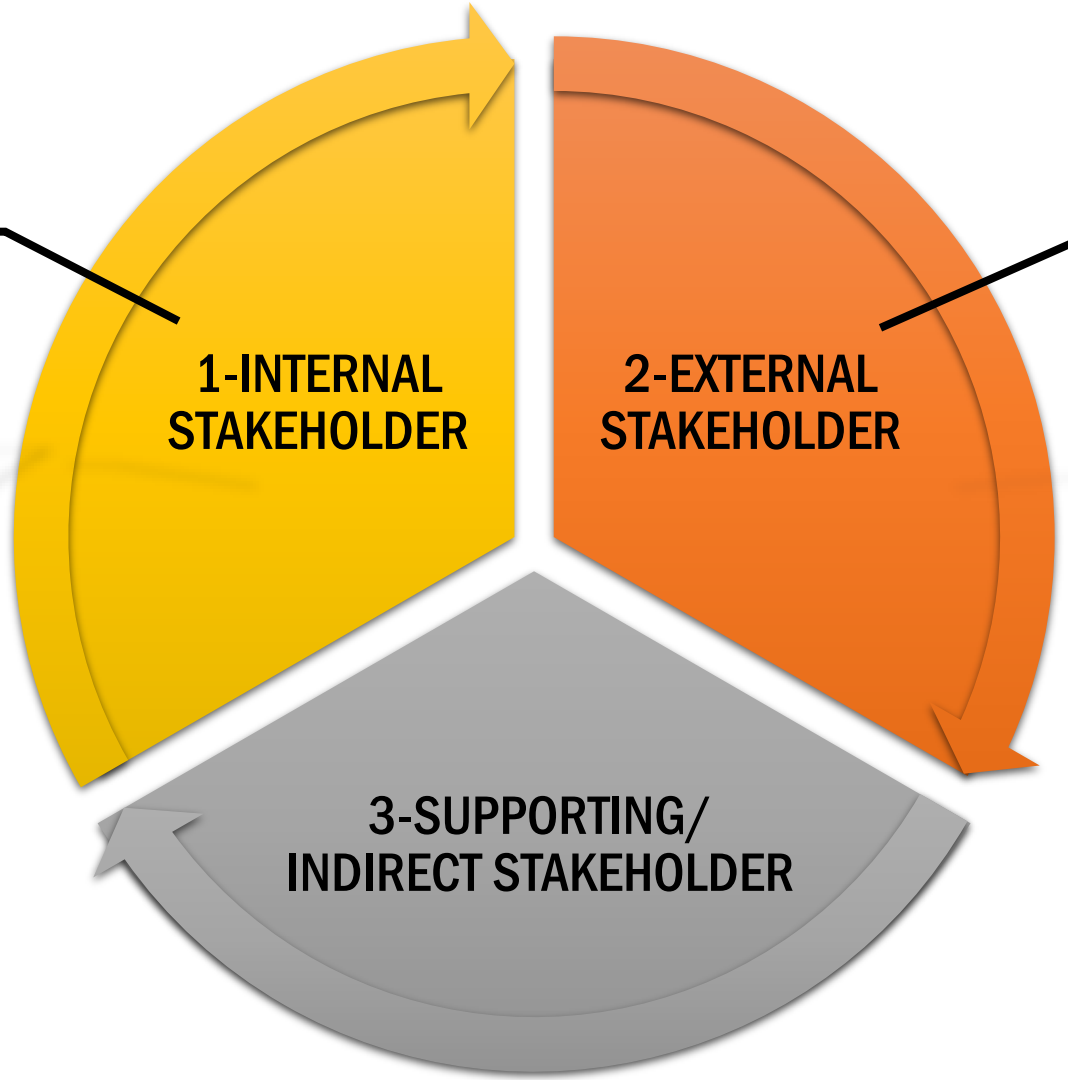
Academic and Practical Contribution

The study contributes to academic literature while offering practical insights that organizations can apply in real-world project scenarios.



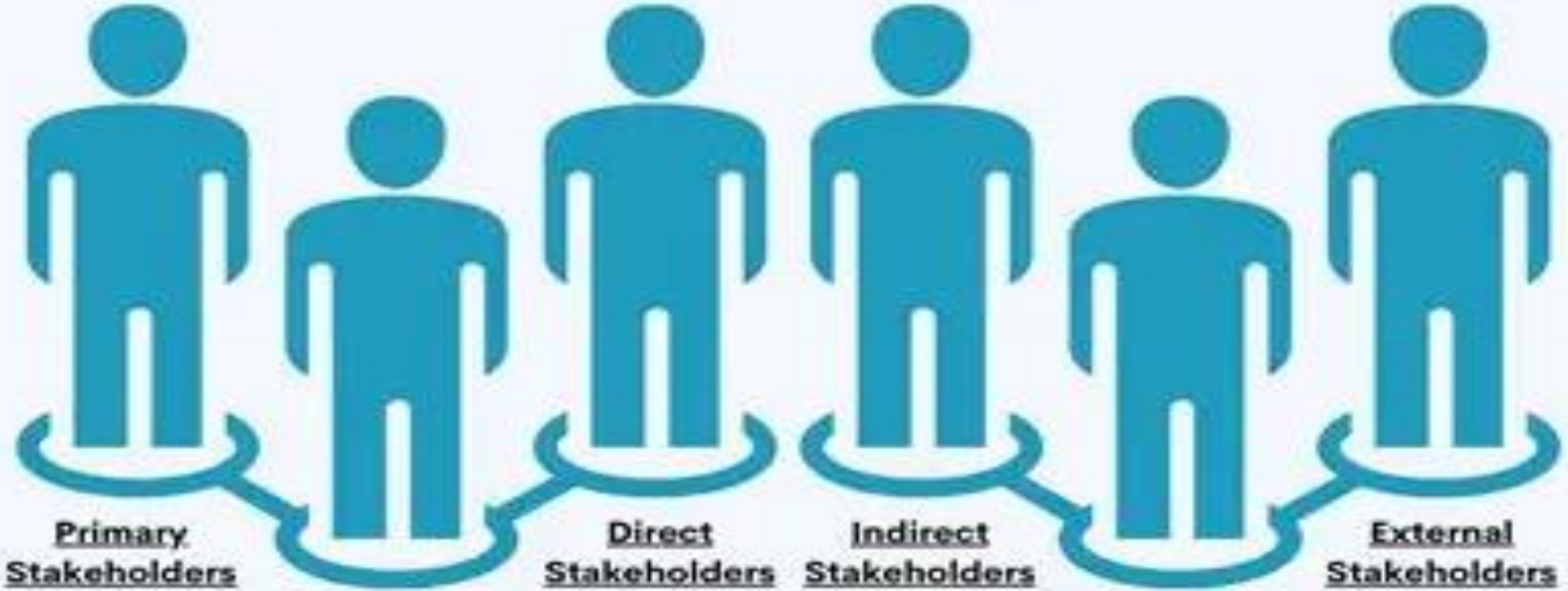
STAKEHOLDER

These stakeholders are directly involved in planning, executing, and delivering the project



These stakeholders influence or are impacted by the project's sustainability and ESG outcomes.

Different Types of Stakeholders



Primary Stakeholders
Customers
Investors
Suppliers

Secondary Stakeholders
Industry Experts
Consultants
Competitors

Direct Stakeholders
Shareholders
Users
Executives

Indirect Stakeholders
Contractors
Vendors
Collaborators

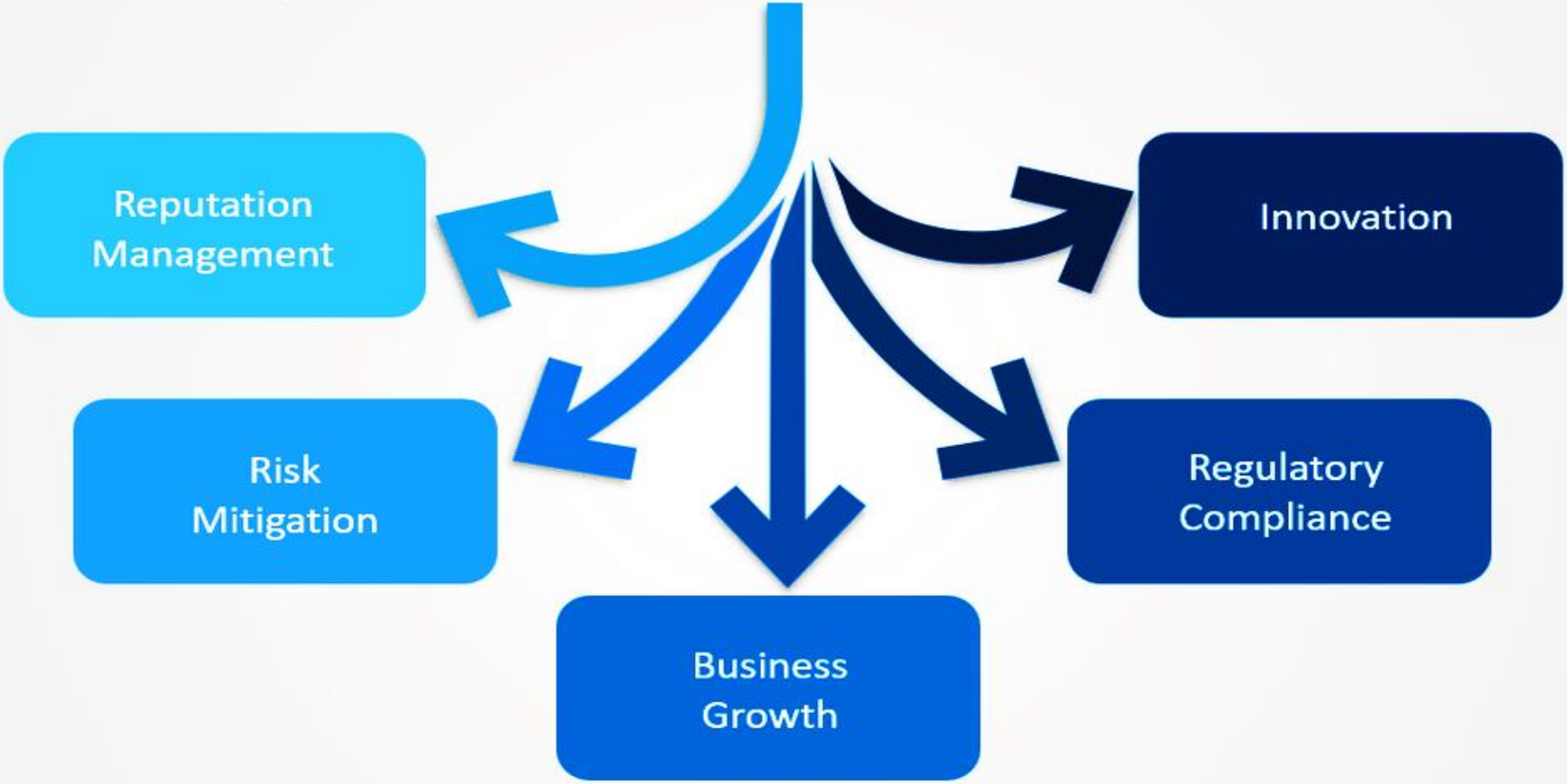
Internal Stakeholders
Team
Executives
Employees

External Stakeholders
Customers
Investors
Suppliers

Stakeholder Map: Who Needs What?



Importance of Stakeholder Relations



Scope of the Project

The project will focus on integrating **Sustainability** and **ESG (Environmental, Social, and Governance)** principles into the entire **project management lifecycle**. This includes:

- **Research & Development**: Identifying best practices, methodologies, and international standards for ESG integration.
- **Policy Development**: Creating frameworks and policies for applying ESG principles to projects (environmental, social, governance).
- **Training & Awareness Programs**: Conducting training for project managers and other stakeholders on the importance and application of ESG principles.
- **Monitoring & Reporting**: Setting up systems to track and report ESG progress throughout the project lifecycle.
- **Stakeholder Engagement**: Communicating ESG objectives and achievements with internal and external stakeholders to ensure alignment and transparency.



Cost Estimation Methods

For the project on Sustainability and ESG Integration in Project Management, based on the detailed breakdown of costs and the complexity of the project, the cost estimation method that has likely been adopted is a combination of multiple techniques, with an emphasis on Bottom-Up Estimating and Parametric Estimating.

➢ Bottom-Up Estimating

This method is the most suitable for a project like Sustainability and ESG Integration, where detailed scope and tasks are involved. The project likely includes several distinct work packages such as:

- Research & Development
- Policy Development
- Training & Awareness Programs
- Monitoring & Reporting
- Stakeholder Engagement

Each of these components has a clearly defined scope that can be broken down into smaller tasks, making Bottom-Up Estimating a highly appropriate method for creating detailed and accurate cost estimates. By estimating costs at the work package level (e.g., how much it will cost to create ESG policies or implement training programs), you can sum these estimates to arrive at a total project cost.

This method allows for a granular view of project costs and ensures accuracy by breaking the project into smaller, manageable components.

Bottom-Up Estimating Example

Work Breakdown Structure (WBS): For example, costs for each phase of the project (research, policy development, training) would be estimated individually (labor, resources, external consultants, software tools), then aggregated to provide a comprehensive total project cost.

➢ Parametric Estimating

Since ESG Integration projects involve some standard costing factors (e.g., cost per unit of training for employees, cost per unit of consultant time, cost per project hour for sustainability tools), Parametric Estimating is also likely to have been applied. This method uses established relationships between variables and costs. For instance:

- **Cost per training hour:** If the cost of a workshop is known (say $500 per session), the cost of training 10 sessions can be estimated.
- **Cost per software tool user license:** If the price for ESG monitoring software is available on a per-user basis, the cost for scaling to the project size can be estimated.

Parametric Estimating Example

Cost per unit of software: If the project requires a specific ESG monitoring tool, and the software vendor charges based on the number of users or licenses, then parametric estimating could be used to project software costs based on the estimated number of users in the project.

➢ Reserve Analysis

In a project like Sustainability and ESG Integration, which often deals with uncertainty and evolving regulations, **Reserve Analysis** is an essential component of the cost estimation. Contingency reserves would be added to address potential unforeseen risks (e.g., new environmental regulations, unexpected costs for external consultations, or additional training requirements). This helps cushion the overall project cost estimate to ensure the project remains viable despite uncertainties.

Adopted Method for This Project

Given the complexity and multi-phase nature of your Sustainability and ESG Integration project, the cost estimation approach is likely a hybrid of the following:

1. Bottom-Up Estimating (for accurate, task-specific cost breakdowns)
2. Parametric Estimating (for estimating costs based on standard variables like training hours, software tools, and consultant fees)
3. Reserve Analysis (to account for unknown risks and uncertainties in the project lifecycle).

These methods combined ensure the cost estimate is both accurate and flexible, accommodating the need for detailed planning and managing potential risks effectively.

Project Cost Breakdown by Phase

Phase	Activity	Cost Range	Total Cost for Phase
Phase 1: Initial Planning & Research (15-20%)	**Market Research:** Identifying current ESG best practices, standards, and regulations.	$15,000 - $30,000	**$30,000 - $100,000**
	Consultancy & Expertise: Hiring external ESG consultants for technical guidance.	$10,000 - $40,000	
	Framework & Tool Development: Creating ESG frameworks, policies, and systems.	$5,000 - $30,000	
Phase 2: Policy Development (10-15%)	**Stakeholder Engagement:** Working with management and departments to design and refine policies.	$10,000 - $25,000	**$20,000 - $60,000**
	Legal Compliance: Aligning policies with regional/national regulations.	$5,000 - $15,000	
	Documentation: Preparing official documentation and internal guides.	$5,000 - $20,000	
Phase 3: Training & Awareness (10-20%)	**Workshops and Training Programs:** Conducting in-house or outsourced workshops for project managers and staff.	$15,000 - $40,000	**$25,000 - $75,000**
	Development of Educational Materials: Designing training materials, online courses, and certifications.	$10,000 - $35,000	
Phase 4: Execution & Monitoring (20-30%)	**ESG Integration:** Implementing sustainability initiatives across projects (e.g., energy-saving, waste management).	$20,000 - $50,000 per project	**$50,000 - $150,000**
	Monitoring Systems: Implementing software tools for monitoring and reporting ESG compliance.	$25,000 - $100,000	
	Data Collection & Analysis: Setting up processes for data collection and regular reporting on ESG targets.	$5,000 - $20,000	
Phase 5: Stakeholder Engagement & Communication (10-15%)	**Public Relations:** Promoting ESG achievements through reports, media, and public campaigns.	$10,000 - $30,000	**$15,000 - $50,000**
	Community Engagement: Engaging with local communities or other external stakeholders regarding ESG practices.	$5,000 - $20,000	
Total Estimated Project Cost			$140,000 - $435,000

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Risk Assessment and Risk Register

The Risk Register identifies potential risks that might occur during the project lifecycle. Below are possible risks and their management strategies:

Risk ID	Risk Description	Probability	Impact	Mitigation Strategy
R1	**Regulatory and Compliance Risk**	High	High	Regular compliance checks, use legal experts to ensure alignment with ESG regulations, and update policies as required by law.
R2	**Resistance to Change**	Medium	High	Implement a comprehensive change management strategy, engage stakeholders early, and communicate benefits of ESG integration.
R3	**Inaccurate ESG Data Reporting**	Medium	High	Invest in reliable data collection and reporting tools, establish regular audits, and train teams to ensure accurate data collection.
R4	**Increased Project Costs Due to Sustainability Requirements**	Medium	Medium	Prepare a detailed cost-benefit analysis showing long-term savings, secure internal buy-in, and phase implementation for sustainability measures.
R5	**Resistance from External Stakeholders**	Low	Medium	Engage with external stakeholders early, share ESG goals, and align the project's outcomes with their expectations and concerns.
R6	**Supply Chain Disruptions Due to ESG Practices**	Medium	High	Diversify suppliers, build strong relationships with ethical suppliers, and monitor supply chain sustainability.
R7	**Lack of Skilled Personnel in ESG Integration**	Low	Medium	Invest in employee training and hire ESG experts for initial phase implementation.
R8	**Technological Challenges in ESG Monitoring Tools**	Medium	Medium	Conduct thorough vendor evaluations, ensure integration of tools with existing systems, and provide proper training for end-users.
R9	**Environmental Uncertainty (e.g., Climate Change Impact)**	Low	High	Incorporate flexible adaptation strategies, use predictive models, and ensure the project's ESG framework is adaptable to changing environmental conditions.

Risk Matrix Analysis

The **Risk Matrix** helps assess the probability and impact of each risk on a scale of 1 to 5. Based on this, you can prioritize actions to mitigate high-priority risks.

Risk	Probability (1-5)	Impact (1-5)	Risk Score	Priority Level
Regulatory and Compliance Risk	5	5	25	**Critical**
Resistance to Change	3	5	15	**High**
Inaccurate ESG Data Reporting	3	5	15	**High**
Increased Project Costs Due to Sustainability	3	4	12	**Moderate**
Resistance from External Stakeholders	2	3	6	**Low**
Supply Chain Disruptions	3	5	15	**High**
Lack of Skilled Personnel	2	4	8	**Moderate**
Technological Challenges	3	4	12	**Moderate**
Environmental Uncertainty	2	5	10	Moderate

Risk Priority Classification

Critical (20-25)

Immediate action is required to mitigate.

High (10-19)

Plan mitigation strategies.

Moderate (5-9)

Monitor and ensure mitigation is in place.

Low (1-4)

Minimal attention required, monitor periodically.

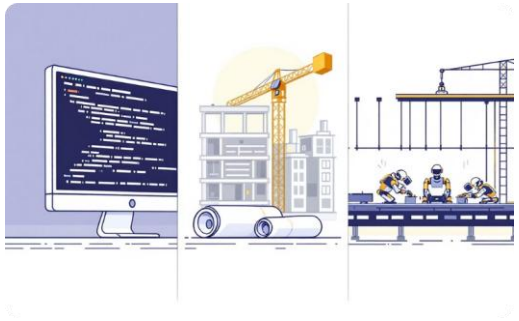
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Complexity Analysis



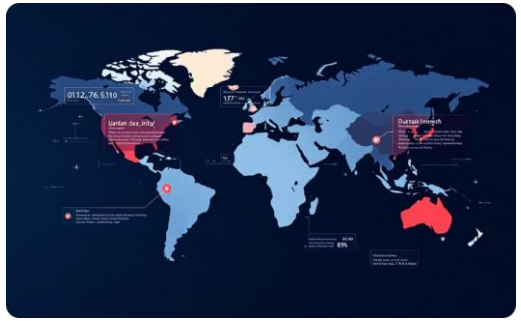
Cross-Departmental Collaboration

ESG integration is complex, requiring input and cooperation from diverse departments. This necessitates clear role definition, regular communication, and effective collaboration tools.



Customization Across Different Projects

ESG principles must adapt to varying project types (IT, construction, manufacturing). Flexible frameworks and guidelines are crucial for effective application.



Global Standards and Reporting

Varying global ESG standards and reporting requirements create alignment challenges. A flexible framework is needed to meet regional needs while upholding core global standards.



Long-Term Commitment

Maintaining consistent ESG integration demands long-term commitment and continuous monitoring. This requires realistic expectations, sustained reporting, and dedicated resources for ongoing improvement.



PM Methodologies for Sustainability & ESG Integration



1. PRiSM (Projects Integrating Sustainable Methods)

Key Features

- Integrates sustainability at every stage.
- Uses the P5 Standard (People, Planet, Prosperity, Process, Products).
- Includes lifecycle impact assessments.
- Reduces environmental & social impacts

Most Likely Used in:

Green projects, construction, government-funded initiatives, ESG-sensitive industries.

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2. Agile + ESG Integration

Key Features

- Adaptive sustainability controls in sprints, story mapping.
- Continuous stakeholder feedback on environmental/social impact.
- Short iterations allow frequent sustainability checks.

Most Likely Used in:

Software development, R&D projects.



3. PMBOK (ISO 21500 & ISO 21502 (Guides to Project Management) – ESG-Aligned

How it Support ESG

- ESG risk assessment in the Risk Management Plan.
- Sustainability KPIs in the Project Charter.
- Environmental/social monitoring under Quality & Stakeholder Engagement.
- Procurement tied to ESG criteria

Most Likely Used in:

Standardized governance & stakeholder focus

Large-scale capital projects, manufacturing, engineering, public sector.



4. Systems Thinking Approach

Key Features

- Environmental impact modelling.
- Life cycle analysis.
- Circular economy-based project planning

Most Likely Used in:

Lifecycle and environmental modeling

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5. Value Management (VM) with ESG Focus

How it supports ESG

- Evaluates environmental & social value, not only cost.
- Identifies opportunities for resource efficiency.
- Encourages long-term sustainable outcomes.

Most Likely Used in:

Optimizing environmental/social/economic value

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6. Six Sigma & Lean with Sustainability

How it supports ESG

- Minimizing waste (environmental focus).
- Process efficiency (economic focus).
- Better working environments (social focus)

Most Likely Used in:

Waste reduction and efficiency aligned with ESG

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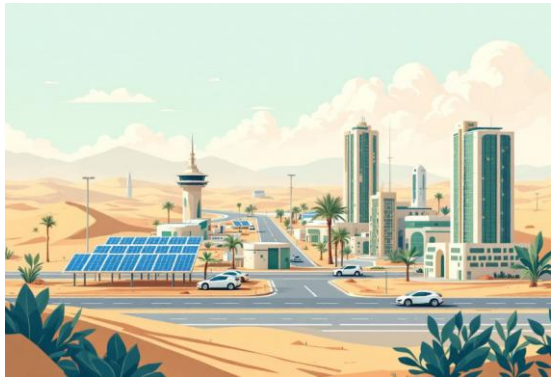
Global Lessons for Project Managers

Project managers around the world can learn from real successes like London 2012 and Masdar City and major failures like BP Deepwater Horizon and the Three Gorges Dam.

These cases clearly show what creates sustainable success and what causes ESG breakdowns in mega-projects.



Project Case Studies | Learning from Successes and Setbacks |



Masdar City | UAE

Masdar City in Abu Dhabi is **one of the world's first planned zero-carbon, eco-friendly cities,** designed **to run almost entirely on solar power and smart technologies**. It features **energy-efficient buildings that stay cool with minimal air-conditioning, electric driverless transport, and systems** that use very little energy, making it a global model for future sustainable urban development.



London 2012 Olympics

The London 2012 Olympic & Paralympic Games set a new standard for **sustainable major events, aiming to create a lasting green legacy**. The Olympic Park was **transformed from a former industrial site into a biodiverse urban green space, utilizing recycled materials extensively & implementing robust waste and water management strategies.**



Three Gorges Dam | China

While **an engineering marvel providing vast hydroelectric power,** the Three Gorges Dam project led to massive population displacement, significant environmental degradation, and increased geological risks. Its immense scale also caused changes in the local ecosystem, impacting biodiversity and creating water quality issues, making it a controversial example of unsustainable development.



BP Deepwater Horizon | Gulf of Mexico

The **Deepwater Horizon oil spill was one of the largest environmental disasters** in history, causing **immense ecological damage to the Gulf of Mexico**. The **failure highlighted severe shortcomings in risk management, safety protocols and emergency response** in mega-projects, resulting in long-term environmental and economic consequences.

Project	Scope	Budget (Planned → Actual)	Timeline (Expected → Actual)
London 2012 Olympics (UK)	Build venues; regenerate East London; deliver sustainable Games; long-term legacy	£2.4B → £9.3B (revised) → **£8.4B actual**	2007–2011 → **Completed on time, Games in 2012**
Masdar City (UAE)	Build zero-carbon eco-city; smart mobility; solar energy; green buildings; clean-tech hub	~$22B → **$18–19.8B** (scaled down)	Complete by 2016 → **Extended to 2030+**
Three Gorges Dam (China)	Build world's largest hydropower dam; improve flood control; improve navigation; resettlement	¥90B (~$12B) → **¥203–249B** (~$30–37B)	1993–2009 → Main dam 2006 (early), full operations by 2012
Deepwater Horizon (USA, 2010)	Safely drill the Macondo Prospect well; meet offshore safety + environmental standards	Standard drilling budget → **$63.4–65B disaster cost**	Routine drilling → **Explosion + 87-day oil spill**

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1. Plan ESG from Day One

Why it matters:

When sustainability is built into the project charter, design, procurement and timeline from the start, long-term value increases and risks drop.

Success - London 2012 Olympics

- Planned sustainability from the very beginning
- Achieved zero waste to landfill, used recycled materials, built reusable venues
- Regenerated East London into lasting community infrastructure

What it teaches:

✓ Early ESG integration leads to operational efficiency, credibility, and long-term community impact.

Failure - Three Gorges Dam

- Social and environmental issues were not considered early
- Resulted in 1.3 million displaced people, cultural loss, and ecosystem disruption

What it teaches:

✗ Delayed or weak ESG planning creates irreversible human and environmental harm.

2. Use Strong Governance & Transparency

Why it matters:

Clear reporting, honest risk assessment and strong oversight prevent accidents and build trust.

Failure - BP Deepwater Horizon

- Safety tests skipped, warnings ignored, poor oversight
- Led to the largest marine oil spill, 11 deaths, $65B in penalties

What it teaches:

✗ Weak governance and hidden problems create catastrophic ESG failures.

Success - London 2012

- Used transparent ESG scorecards
- Published sustainability results for public accountability

What it teaches:

✓ Transparency increases performance and protects project integrity.

3. Balance People, Planet & Profit (Social + Environmental Integration)

Why it matters:

A project is not sustainable if it delivers technical success but hurts communities or ecosystems.

Failure - Three Gorges Dam

- Clean energy achieved, but massive social and ecological damage

What it teaches:

✗ Technical success alone is not ESG success. ✗ Social impact must be weighed equally with environmental and economic benefits.

Success - Masdar City

- Built a low-carbon, energy-efficient city with smart transport
- Created jobs and attracted global clean-tech companies

What it teaches:

✓ Balanced environmental and social planning creates resilient cities and long-term economic value.

4. Never Compromise Safety or Environmental Controls

Why it matters:

Shortcuts to save time or money always cost more later - financially, legally and ethically.

Failure - BP Deepwater Horizon

- Safety rules bypassed to save costs
- Led to long-term ecological and financial damage

What it teaches:

✗ Rushing or ignoring risks destroys reputations, budgets, and ecosystems. ✗ Safety and environmental controls must be non-negotiable.

Success - London 2012 Olympics

- Carried out **strict safety protocols** across all construction zones
- Zero major safety incidents despite massive workforce and tight schedule
- Robust environmental controls kept **99% of waste recycled** and avoided pollution

What it teaches:

✓ Strong safety culture + strict environmental practices can deliver large-scale projects **on time, safely, and sustainably**.

5. Engage Stakeholders Continuously (Community, Suppliers, Regulators)

Why it matters:

Projects succeed when communities are heard, suppliers are aligned and regulators are involved early.

Failure - Three Gorges Dam

- Poor community engagement → mass displacement, long-term social resentment

What it teaches:

✗ Ignoring people leads to protests, delays, and loss of legitimacy.

Success - London 2012 & Masdar City

- Transparent communication
- Community input & green supplier requirements

What it teaches:

✓ Collaboration reduces conflict and increases project acceptance.

Takeaway

Successful projects (London 2012, Masdar City) prove that:

ESG works when planned early, governed strongly and integrated with innovation.

Failed projects (BP Deepwater Horizon, Three Gorges Dam) show that:

Ignoring safety, communities, or environmental risks leads to financial loss, human harm and long-term damage.

For every project manager:

Sustainability succeeds when **ESG is proactive, transparent and people-centered** and fails when shortcuts or weak governance take over.



Emerging Trends in Sustainability & ESG Integration in Project Management

ESG as a Mandatory Requirement

- ESG reporting is becoming legally required, not optional.
- Governments and investors are now forcing companies to adopt ESG, especially in large infrastructure and public projects.
- Projects must show clear evidence of environmental, social, and governance performance.

Digital ESG Reporting & AI Tools

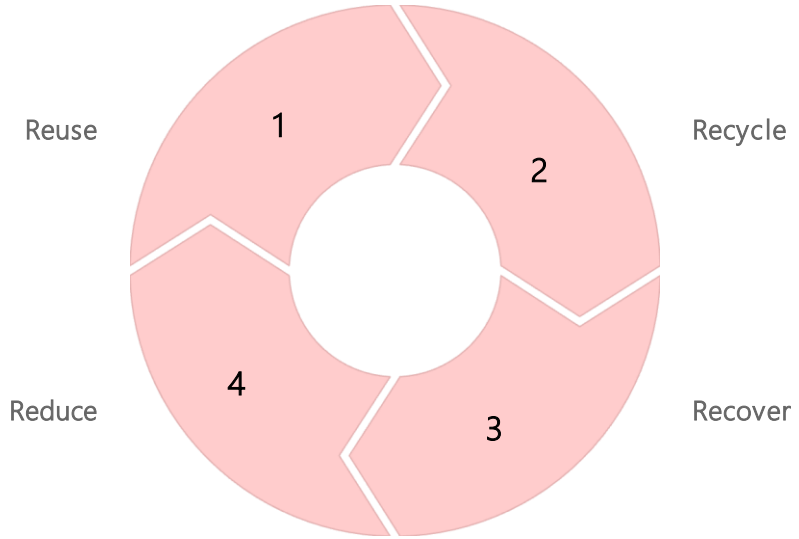
- Companies now use AI, automation, and digital dashboards to track:
  - Carbon emissions
  - Waste management
  - Worker safety
  - Social impact
- AI platforms help project managers predict ESG risks and improve compliance.

Green Procurement (Sustainable Supply Chain)

- Selecting suppliers based on ESG performance.
- Companies buy:
  - Eco-friendly materials
  - Energy-efficient equipment
  - Certified sustainable products
- Suppliers must meet ISO 14001, ISO 26000, or ESG ratings.

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Circular Economy Practices



Instead of "use → throw away", projects now follow this circular flow.

Examples:

- Reusing construction materials
- Designing products for long-term reusability
- Reducing landfill waste

Reduces project costs and environmental impact.

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Social Sustainability as a Core Requirement



Worker Mental Health & Well-being



Local Community Benefits



Diversity & Inclusion



Community Engagement Sessions

ESG Risk Management Frameworks

ESG risk assessments are integrated into early planning phases, identifying potential challenges such as:

Climate Risks

Supply Chain Disruptions

Social Conflicts

Legal Risks

Reputational Damage

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Green Financing & ESG-Based Funding

Lower Interest Rates

Banks and investors offer reduced interest rates for projects demonstrating strong ESG performance and compliance.

Limited Funding Access

Projects that lack ESG compliance or integration may encounter significant challenges in securing necessary financing.

Use of International Sustainability Standards

Growing adoption of key international standards helps managers track ESG KPIs clearly and comparably:

1

GRI Standards

2

SASB

3

ISO 14001, 26000, 50001

4

IFRS Sustainability Standards

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Renewable & Smart Technologies in Projects

Projects now inegrate these advanced technologies, moving them from luxury to standard:

Smart Meters

Monitors energy consumption

IoT-based Energy Monitoring

Uses interconnected sensors for real-time tracking

Green Buildings

LEED certification for sustainable design

Solar Power Integration

Harnessing renewable energy

Energy-efficient HVAC Systems

Optimized climate control

Smart Waste Systems

Efficient waste collection and management

These technologies are quickly becoming standard, not luxury, in modern projects.

Human-Centric Project Design

Modern projects prioritize improving user experience and community quality of life through key design principles:

Accessible Infrastructure

Safe Public Spaces

Better Worker Conditions

Gender-Sensitive Design

People-first design is now a critical sustainability requirement.

Top ESG Success Stories

1. Bosco Verticale (Vertical Forest) - Milan, Italy

Problem

- Lack of green spaces in dense cities
- Rising urban heat and pollution
- Biodiversity loss

Solution

- **Built residential towers with 900+ trees and 20,000+ plants on balconies**
- Vegetation acts as an air filter and temperature moderator

Results / Impact

- **30 tons of CO_2 absorbed annually**
- Reduced heat-island effect
- Improved biodiversity
- Better resident well-being



2. Transport Sector - UPS ORION Route Optimization (USA)

Problem

- UPS delivery trucks burning massive fuel with high carbon emissions, inefficient routes increasing costs and delivery time

Solution

- Developed ORION (On-Road Integrated Optimization & Navigation), **an AI algorithm to optimize delivery routes, analyzes 200,000+ route options per driver daily**

Results / Impact

- **Reduced 100 million miles driven annually, saved 10 million gallons of fuel per year, prevented 100,000 metric tons of CO_2 emissions yearly, improved delivery efficiency**



3. Manufacturing Sector - Unilever Sustainable Packaging (Global)



Problem

- Consumer goods industry generates significant plastic waste, reliance on non-sustainable sourcing, impact on ecosystems and resource depletion

Solution

- Pledged to halve plastic use and increase recycled content, committed to 100% sustainably sourced key agricultural raw materials, invested in innovative packaging solutions

Results/ Impact

- **Reduced plastic footprint by 13% since 2019, achieved 67% s**ustainably sourced agricultural raw materials, enhanced brand reputation, influenced broader industry shift

Key Takeaways

1	ESG integration drives environmental, social, and financial benefits in projects
2	Digital tools and smart technologies enhance efficiency and reduce emissions
3	Circular economy and resource optimization minimize waste and support sustainability
4	Social initiatives strengthen workforce diversity, community impact, and inclusion
5	Sustainable projects create competitive advantage, stakeholder trust, and long-term value

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